Level 3

Notes:

1. **'id' in df1 is the same as 'question_id' in df2.**
2. The question_id is the same for both questions in En and Fr for a given test_design_id. When test_design_id changes, so does the question_id.
3. Each specific question is specified by question_id, lang or alternatively by the response_id which combines the question_id, the language, and the subpart number.
    a. For example, the below shows the unique response_ids corresponding to question_id = 18712 in lang = en. All of them have **18712** (the question id), **e11** (code corresponding to en, for fr, it is e28), and the index for the subpart number (r0, r1, etc.)

```
18712 en ['i18712.e11.r0' 'i18712.e11.r1' 'i18712.e11.r2' 'i18712.e11.r3'
 'i18712.e11.r4' 'i18712.e11.r5' 'i18712.e11.r6' 'i18712.e11.r7'
 'i18712.e11.r8']
```

4. **Issue:** I find that the response ids and question_ids which correspond to Question_15 and Question_17 in the questionnaire (df1) are simply not there in the responses (df2). In fact, the response csv file (df2) only contains response_ids and question_ids corresponding to Question 1 from df1 i.e.

```
5. df2['response_id'].unique() # = array(['i16961.e11.r0', 'i16961.e11.r1',
   'i16961.e11.r2', 'i16961.e11.r3'])
6. df2['question_id'].unique() # = array([16961])
```

7. However, the response_set_ids hint towards the response_ids of other questions. We can split response_set_id at the full stops ("."). This is done as follows:

If response_set_id = r111871274.i16961.e11.r0, then it is splitted into [r11**18712**74, i16961, e11, r0].

From the first element, we have question_id = **18712**. And then the language code ('e11' or 'e28'') should be taken from the lang variable instead of the above list because there are records where the language code in response_set_id is misaligned with the lang variable: For example:

```
r107781013.i16961.e11.r2,...,fr
```

However, whether the 'r0' above or the last entry of the response_set_id can't be trusted as it is repeated in the responses dataframe (r0,r1,r2,r3,r0,r1,r2,r3,....). Furthermore, the response_id in the questionnaire was uniquely defined by the test_version_id and the question_id, however in df2 all question_ids are set to 16961. So it seems impossible to me as of yet to figure out whether a response_set_id refers correctly to r0,r1,r2,.., r7 or r8.

But let's say we take the codes r0, r1, r2, r3 at the end of response_set_id seriously. Then I can recreate a new response_id for each entry in df2 and inner join it with df1. This is shown below for Question 15 and Question 17 with the code. **One anomaly is that the same**

response, say C, is associated with all of r0, r1, and r2 which shouldn't be the case according to df1 where all separate options A,B, C, D are associated with r0,r1,r2,r3.

	id	response_id_x	response_set_id	label	is_correct	response	response_text	score	is_nr
300	18712	i18712.e11.r0	r118712062.i16961.e11.r0	Question 15	0	C	NaN	0.0	0
301	18712	i18712.e11.r0	r118712290.i16961.e11.r0	Question 15	0	C	NaN	0.0	0
302	18712	i18712.e11.r0	r118712752.i16961.e11.r0	Question 15	0	C	NaN	0.0	0
303	18712	i18712.e11.r1	r118712062.i16961.e11.r1	Question 15	0	C	NaN	0.0	0
304	18712	i18712.e11.r1	r118712290.i16961.e11.r1	Question 15	0	C	NaN	0.0	0
305	18712	i18712.e11.r1	r118712752.i16961.e11.r1	Question 15	0	C	NaN	0.0	0
306	18712	i18712.e11.r2	r118712062.i16961.e11.r2	Question 15	0	C	NaN	0.0	0
307	18712	i18712.e11.r2	r118712290.i16961.e11.r2	Question 15	0	A	NaN	0.0	0
308	18712	i18712.e11.r2	r118712752.i16961.e11.r2	Question 15	0	C	NaN	0.0	0
309	18712	i18712.e11.r3	r118712062.i16961.e11.r3	Question 15	0	C	NaN	0.0	0
310	18712	i18712.e11.r3	r118712290.i16961.e11.r3	Question 15	0	C	NaN	0.0	0
311	18712	i18712.e11.r3	r118712752.i16961.e11.r3	Question 15	0	B	NaN	0.0	0
380	18759	i18759.e11.r0	r118759011.i16961.e11.r0	Question 15	0	B	NaN	0.0	0
381	18759	i18759.e11.r0	r118759771.i16961.e11.r0	Question 15	0	C	NaN	0.0	0
382	18759	i18759.e11.r0	r118759935.i16961.e11.r0	Question 15	0	C	NaN	0.0	0
383	18759	i18759.e11.r1	r118759011.i16961.e11.r1	Question 15	0	B	NaN	0.0	0
384	18759	i18759.e11.r1	r118759771.i16961.e11.r1	Question 15	0	A	NaN	0.0	0
385	18759	i18759.e11.r1	r118759935.i16961.e11.r1	Question 15	0	C	NaN	0.0	0
386	18759	i18759.e11.r2	r118759011.i16961.e11.r2	Question 15	0	C	NaN	0.0	0
387	18759	i18759.e11.r2	r118759771.i16961.e11.r2	Question 15	0	C	NaN	0.0	0
388	18759	i18759.e11.r2	r118759935.i16961.e11.r2	Question 15	0	A	NaN	0.0	0
389	18759	i18759.e11.r3	r118759011.i16961.e11.r3	Question 15	0	C	NaN	0.0	0
390	18759	i18759.e11.r3	r118759771.i16961.e11.r3	Question 15	0	B	NaN	0.0	0
391	18759	i18759.e11.r3	r118759935.i16961.e11.r3	Question 15	0	C	NaN	0.0	0
392	18759	i18759.e28.r0	r118759261.i16961.e11.r0	Question 15	0	C	NaN	0.0	0
393	18759	i18759.e28.r1	r118759261.i16961.e11.r1	Question 15	0	C	NaN	0.0	0
394	18759	i18759.e28.r2	r118759261.i16961.e11.r2	Question 15	0	B	NaN	0.0	0
395	18759	i18759.e28.r3	r118759261.i16961.e11.r3	Question 15	0	C	NaN	0.0	0

	id	response_id_x	response_set_id	label	is_correct	response	response_text	score	is_nr
316	18715	i18715.e11.r0	r118715076.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
317	18715	i18715.e11.r0	r118715089.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
318	18715	i18715.e11.r0	r118715124.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
319	18715	i18715.e11.r0	r118715182.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
320	18715	i18715.e11.r0	r118715798.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
321	18715	i18715.e11.r1	r118715076.i16961.e11.r1	Question 17	0	B	NaN	0.0	0
322	18715	i18715.e11.r1	r118715089.i16961.e11.r1	Question 17	0	B	NaN	0.0	0
323	18715	i18715.e11.r1	r118715124.i16961.e11.r1	Question 17	0	C	NaN	0.0	0
324	18715	i18715.e11.r1	r118715182.i16961.e11.r1	Question 17	0	B	NaN	0.0	0
325	18715	i18715.e11.r1	r118715798.i16961.e11.r1	Question 17	0	C	NaN	0.0	0
326	18715	i18715.e11.r2	r118715076.i16961.e11.r2	Question 17	0	B	NaN	0.0	0
327	18715	i18715.e11.r2	r118715089.i16961.e11.r2	Question 17	0	C	NaN	0.0	0
328	18715	i18715.e11.r2	r118715124.i16961.e11.r2	Question 17	0	A	NaN	0.0	0
329	18715	i18715.e11.r2	r118715182.i16961.e11.r2	Question 17	0	C	NaN	0.0	0
330	18715	i18715.e11.r2	r118715798.i16961.e11.r2	Question 17	0	C	NaN	0.0	0
402	18761	i18761.e11.r0	r118761380.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
403	18761	i18761.e11.r0	r118761390.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
404	18761	i18761.e11.r0	r118761469.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
405	18761	i18761.e11.r0	r118761846.i16961.e11.r0	Question 17	0	C	NaN	0.0	0
406	18761	i18761.e11.r1	r118761380.i16961.e11.r1	Question 17	0	C	NaN	0.0	0
407	18761	i18761.e11.r1	r118761390.i16961.e11.r1	Question 17	0	C	NaN	0.0	0
408	18761	i18761.e11.r1	r118761469.i16961.e11.r1	Question 17	0	C	NaN	0.0	0
409	18761	i18761.e11.r1	r118761846.i16961.e11.r1	Question 17	0	C	NaN	0.0	0
410	18761	i18761.e11.r2	r118761380.i16961.e11.r2	Question 17	0	B	NaN	0.0	0
411	18761	i18761.e11.r2	r118761390.i16961.e11.r2	Question 17	0	C	NaN	0.0	0
412	18761	i18761.e11.r2	r118761469.i16961.e11.r2	Question 17	0	C	NaN	0.0	0
413	18761	i18761.e11.r2	r118761846.i16961.e11.r2	Question 17	0	C	NaN	0.0	0

Code

```
# load the data using pandas
import pandas as pd
import numpy as np
pd.set_option('display.max_colwidth', None)
df1 = pd.read_csv("QuestionnaireQuestions-202209061907.csv")
df2 = pd.read_csv("SubmittedQuestionResponses-202209061913.csv")
pd.set_option('display.max_rows', 10)
from IPython.display import display


new_values = [] # holds new response ids which I recreate from the response_set_id
for row in df2.itertuples(index=False):
   splitted = row.response_set_id.split('.')
   question_id = splitted[0][2:7]
   if row.lang == 'en':
       lang_code = '.e11.'
   else:
       lang_code = '.e28.'
   response_id = 'i'+question_id+lang_code+splitted[-1] # I take the option code
(r0,r1,..) code which is given in response_set_id
   new_values.append(response_id)

df2['n_response_id'] = new_values
df2 = df2[['response_set_id','n_response_id', 'response_id', 'is_correct',
                                'response', 'response_text', 'score', 'is_nr',

'question_label','is_questionnaire',"is_excluded","is_invalid","is_operational","is
_sample"]]

merged = df1.merge(df2, left_on='response_id', right_on='n_response_id',
how='inner')
merged = merged[['id', 'response_id_x', 'response_set_id','label',
      'is_correct', 'response', 'response_text', 'score', 'is_nr']]
# merged[merged['label']=='Question 15'] # less than total in df1 because not
everything has a response
merged = merged[merged['label']=='Question 17']

print(merged.to_string())
```

Hence, I am not able to join these two dataframes correctly yet or there are no other anomalies other than the one mentioned.